

Alexander Dockhorn | Curriculum Vitae

University of Southern Denmark, SDU Metaverse Lab
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Education

Otto von Guericke University <i>PhD, Final Grade: Summa Cum Laude</i>	Magdeburg 2016–2020
Otto von Guericke University <i>M.Sc. Computer Science, Final Grade: 1.0</i>	Magdeburg 2014–2015
University of Abertay <i>Term abroad</i>	Dundee 2012–2013
Otto von Guericke University <i>B.Sc. Computer Science, Final Grade: 1.2</i>	Magdeburg 2010–2014

Working Experience

University of Southern Denmark, SDU Metaverse Lab <i>Associate Professor</i> - Artificial Intelligence - Applications in Health, Robotics, and Serious Games	Odense since 2025
Leibniz University Hannover, Institute for Information Processing <i>Juniorprofessor</i> - Learning Abstractions for Complex Domains - Explainable Machine Learning and Causality	Hannover 2022–2025
Otto von Guericke University Magdeburg <i>Postdoctoral Research Associate</i> - Multi-Objective Optimization and Decision-Making - Bayesian Networks	Magdeburg 2021–2022
Queen Mary University of London, Game AI Research Group <i>Postdoctoral Research Associate</i> - Decision-making in Exponentially Growing Decision Spaces - State and Action Abstractions for Evolutionary Search Algorithms	London 2020–2021
Otto von Guericke University Magdeburg, University Bremen and Salzgitter AG <i>Research Project</i> - Time Series Analysis - Forecasting Demand	Salzgitter 2019
Otto von Guericke University Magdeburg, Faculty Computer Science, CI Group, Supervisor Prof. Dr. Rudolf Kruse <i>PhD Student, Research and Teaching Associate</i>	Magdeburg 2016–2019

ISC Gebhardt*Working Student*

- Forecasting Demand
- Bayesian Modeling

Wolfsburg

2014

SAP Innovation Center*Bachelor Internship*

- Medical Data Analysis for Cancer Treatment
- Bayesian Modeling

Potsdam

2013–2014

British Telecom Research*Research Internship*

- Frequent Item Set Mining
- Correlation and Causation Analysis

Ipswich

2013

Publications

Total Number of Publications: 82, **Number of Peer-reviewed Publications: 73**
Number of First Authorships: 30, **Number of Corresponding Authorships: 32**
h-Index: 16, i10-Index: 25, Total Citations: 1029 (According to Google Scholar, 09/29/2026)

Books and Bookchapters (7)

Apeldoorn, D. (Ed.), **Dockhorn, A.** (Ed.), & Panholzer, T. (Ed.). (2026). AbstractSwarm Multi-Agent Modeling and Simulation: 20 Years of Logistics with Agents and Stations. SpringerBriefs in Computer Science, Springer Nature Singapore. (to be published)

Borgelt, C., **Dockhorn, A.**, & Kruse, R. (2026). "Unsicheres und unscharfes Wissen". Handbuch der Künstlichen Intelligenz, edited by Tanya Braun, Eyke Hüllermeier, Ute Schmid and Günther Görz, De Gruyter Oldenbourg, 2026, pp. 247-288.

Dockhorn, A., & Kruse, R. (2025). An Overview of Reinforcement Learning Algorithms for Causal Discovery. In N. Daras, A. Fytopoulos, P. Pardalos (Eds.), Handbook of Artificial Intelligence and Machine Learning in Decision Making (to be published), Springer

Dockhorn, A., & Kruse, R. (2023). State and Action Abstraction for Search and Reinforcement Learning Algorithms. In Y. P. Kondratenko, V. Kreinovich, W. Pedrycz, A. A. Chikrii, A. M. Gil Lafuente (Eds.), Artificial Intelligence in Control and Decision-making Systems. Springer.

Dockhorn, A.. (2021). Vorhersagebasierte Suche für autonomes Spielen. In: Hölldobler, S. (Hrsg.), Ausgezeichnete Informatikdissertationen 2020. Bonn: Gesellschaft für Informatik e.V.. (pp. 69-78).

Dockhorn, A., & Kruse, R. (2021). Balancing Exploration And Exploitation in Forward Model Learning. In V. Sgurev, V. Jotsov and J. Kacprzyk (Eds.), Advances in Intelligent Systems Research and Innovation. Elsevier.

Dockhorn, A., Saxton, C., & Kruse, R. (2020). Association Rule Mining for Unknown Video Games. In M.-J. Lesot and C. Marsala (Eds.), Fuzzy Approaches for Soft Computing and Approximate Reasoning: Theories and Applications, (pp. 257-270). Studies in Fuzziness and Soft Computing, Springer Cham.

Journal Papers (17)

Fuchs, R., Droste, J., & **Dockhorn, A.** (2026). Gender Discrimination in Competitive Online Games – Perceptions, Rumination, and Moderation, *IEEE Transactions on Games*, (pp. 1-11).

Apeldoorn, D., Jenn, A., **Dockhorn, A.**, Wollstädter, J., Linkerhägner, F., & Panholzer, T. (2026). Toward

the Optimization of General Logistics Scenarios With Multi-Agent Simulations: Results of the AbstractSwarm Multi-Agent Logistics Competition With Application in Hospital Logistics, *IEEE Access*, vol. 14, (pp. 75831-75848).

Schmöcker, R., Schnell, C., & **Dockhorn, A.** (2026). Investigating Scale-Independent UCT Exploration Factor Strategies, *IEEE Transactions on Games*, vol. 18, no. 1, (pp. 211-216).

Schmöcker, R., **Dockhorn, A.**, & Rosenhahn, B. (2025). Accelerating Probabilistic Planning Research: High-Speed Implementations of Stochastic Markov Decision Processes and Their Catalogization, *IEEE Access*, vol. 13, (pp. 193430-193461).

Eberhardinger, M., Goodman, J., **Dockhorn, A.**, Perez-Liebana, D., Gaina, R. D. Gaina, Cakmak, D., Maghsudi, S., & Lucas, S.. From Code to Play: Benchmarking Program Search for Games Using Large Language Models, *IEEE Transactions on Games* (to be published)

Schmöcker, R., and **Dockhorn, A.** (2025). Cascade - A sequential ensemble method for continuous control tasks, *Reinforcement Learning Journal*, (pp. 399-411).

Yang, C.-H., Liao, Y.-S., Huang, C.-R., and **Dockhorn, A.** (2025). Explaining Neural Style Transfer, *IEEE Computational Intelligence Magazine*, vol. 20, no. 3, (pp. 75-76).

Schmöcker, R., **Dockhorn, A.** (2025). A survey of non-learning-based abstractions for sequential decision-making, *IEEE Access*, (pp. 1-16).

Mittenentzwei, S., Garrison, L., Budich, B., Lawonn, K., **Dockhorn, A.**, Preim, B., Meuschke, M. (2025). AI-based character generation for disease stories: A case study using epidemiological data to highlight preventable risk factors, *i-com*.

Xu, L., **Dockhorn, A.** & Perez-Liebana, D. (2023). Elastic Monte Carlo Tree Search, *IEEE Transactions on Games*, vol. 15, no. 4, (pp. 527-537).

Dockhorn, A., & Lucas, S. (2022). Choosing Representation, Mutation, and Crossover in Genetic Algorithms, *IEEE Computational Intelligence Magazine*, vol. 17, no. 4, (pp. 52-53).

Dockhorn, A., Kirst, M., Mostaghim, S., Wieczorek, M., & Zille H. (2022). Evolutionary Algorithm for Parameter Optimization of Context Steering Agents, *IEEE Transactions on Games*, (pp. 1-10).

Dockhorn, A., & Kruse, R. (2021). Modellheuristiken für effizientes Forward Model Learning. *at - Automatisierungstechnik*, vol. 69, no. 10, (pp. 848-857).

Dockhorn, A., & Kruse, R. (2021). Fuzzy Modeling in Game AI. *TWMS Journal of Pure and Applied Mathematics*, vol. 12, no. 1, (pp.54-68).

Apeldoorn, D., & **Dockhorn, A.** (2021). Exception-Tolerant Hierarchical Knowledge Bases for Forward Model Learning. *IEEE Transactions on Games*, (pp. 1-14).

Dockhorn, A., & Kruse, R. (2020). Predicting Cards Using a Fuzzy Multiset Clustering of Decks. *International Journal of Computational Intelligence Systems*, 13(1), (pp. 1207-1217). Atlantis Press

Held, P., **Dockhorn, A.**, & Kruse, R. (2015). On Merging and Dividing Social Graphs. *Journal of Artificial Intelligence and Soft Computing Research*, 5(1), (pp. 23-49).

Conference Papers (45).....

Melhart, D., **Dockhorn, A.**, Drachen, A., and Elmoose Andersen, M. (2026). AURA: Automated Analysis and Reporting of Therapeutically Applied Table-top Role-Playing Games, *FDG '26: Proceedings of the 21st International Conference on the Foundations of Digital Games*, ACM, pp. 1-15

Gerbeaud, A., Lorcet, L., Campan, L., **Dockhorn, A.**, Wolf, F., Lherbier, A., and Addoum, M. (2026). Large Language Models for Behavior Trees Generation in Unreal Engine, *FDG '26: Proceedings of the 21st International Conference on the Foundations of Digital Games*, ACM, pp. 1-8

Schmöcker, R., **Dockhorn, A.**, Rosenhahn, B. (2026). Grouping Nodes with known Value Differences: A

lossless UCT-based Abstraction Algorithm, *14th International Conference on Learning Representations*, pp. 5440-5459

Schmöcker, R., **Dockhorn, A.**, Rosenhahn, B. (2026). AUPO - Abstracted until proven otherwise: A reward distribution based abstraction algorithm, *Proceedings of the Thirty-Sixth International Conference on Automated Planning and Scheduling, AAAI*, vol. 36, no. 1, pp. 256-265

Droste, J., Fuchs, R., Deters, H., Obaidi, M., **Dockhorn, A.**, Schneider, K. (2026). Immersive and Enjoyable Explanations - On Distinct Explainability Requirements in Games, *Requirements Engineering: Foundation for Software Quality, 32nd International Working Conference*, LNCS, Springer Nature, pp. 3-19

Jiwatode, M., **Dockhorn, A.**, Rosenhahn, B. (2026). From Gameplay Traces to Game Mechanics: Causal Induction with Large Language Models, *Pattern Recognition (ICPR 2026)*, LNCS, Springer Nature, pp. 1-16

Fuchs, R., Rupp, F., Bertram, T., Eckert, K., **Dockhorn, A.** (2026). Toxicity in Twitch Chats: An LLM-Based Analysis Across Gaming Communities, *Proceedings of the IEEE Conference on Games 2026*, IEEE (to be published)

Jiwatode, M., Rosenhahn, B., **Dockhorn, A.** (2026). Compiling VGDL into Causal Models, *Proceedings of the IEEE Conference on Games 2026*, IEEE (to be published)

Jiwatode, M., Fuchs, R., Schmöcker, R., Rosenhahn, B., **Dockhorn, A.** (2026). Spatial Reasoning in LLM Game Agents: Impact of Causal Context and Multi-Step Planning, *Proceedings of the IEEE Conference on Games 2026*, IEEE (to be published)

Lashmet, S., and **Dockhorn, A.** (2025). Training a Reinforcement Learning Agent for Tales of Tribute, *Proceedings of the IEEE Conference on Games 2025*, IEEE, pp. 1-4.

Ronge, R., Albrecht, E., Woiwode, D., **Dockhorn, A.** (2025). Generating Ensembles of Search Policies to Solve Baba is You Levels, *Proceedings of the IEEE Conference on Games 2025*, IEEE, pp. 1-8

Fuchs, R., Droste, J., and **Dockhorn, A.** (2025) How do Players Perceive Gender Discrimination? On the Differences of Harassment in Online Games, *Proceedings of the IEEE Conference on Games 2025*, IEEE, pp. 1-8 🏆 **Best Student Paper Award**

Schmöcker, R., and **Dockhorn, A.** (2025). Time-critical and confidence-based abstraction dropping methods, *Proceedings of the IEEE Conference on Games 2025*, IEEE, pp. 1-8

Xu, L., Liu, Z., **Dockhorn, A.**, Perez-Liebana, D., Wang, J., Song, L., Bian, J. (2024). Higher Replay Ratio Empowers Sample-Efficient Multi-Agent Reinforcement Learning, *Proceedings of the IEEE Conference on Games 2024*, IEEE, pp. 1-8

Xu, L. Perez-Liebana, D., **Dockhorn, A.** (2024). Strategy Game-Playing with Size-Constrained State Abstraction, *Proceedings of the IEEE Conference on Games 2024*, IEEE, pp. 1-8

Nübel, C., **Dockhorn, A.**, Mostaghim, S. (2024). Match Point AI: A Novel AI Framework for Evaluating Data-Driven Tennis Strategies, *Proceedings of the Conference on Games 2024*, IEEE, pp. 1-4

Oguz, M. K., **Dockhorn, A.** (2024). Markov Senior - Learning Markov Junior Grammars to Generate User-specified Content, *Proceedings of the IEEE Conference on Games 2024*, IEEE, pp. 1-8

Jiwatode, M., Schlecht, L., **Dockhorn, A.** (2024). Online Optimization of Curriculum Learning Schedules using Evolutionary Optimization, *Proceedings of the Conference on Games 2024*, IEEE, pp. 1-8

Fuchs, R., Gieseke, R., **Dockhorn, A.** (2024). Personalized Dynamic Difficulty Adjustment - Imitation Learning Meets Reinforcement Learning, *Proceedings of the IEEE Conference on Games 2024*, IEEE, pp. 1-2

Apeldoorn, D., **Dockhorn, A.**, Panholzer T. (2024). AbstractSwarm Multi-Agent Logistics Competition: Multi-Agent Collaboration for Improving A Priori Unknown Logistics Scenarios, *Proceedings of the Genetic and Evolutionary Computation Conference Companion, Association for Computing Machinery*, p. 1-2, Melbourne, VIC, Australia

Mohan, A., Benjamins, C, Wienecke, K, **Dockhorn, A.** & Lindauer, M. (2023). Extended Abstract: AutoRL

Hyperparameter Landscapes, *Second International Conference on Automated Machine Learning*, (pp. 1-6), PMLR.

Olson, C., Wagner, L., & **Dockhorn, A.** (2023). Evolutionary Optimization of Baba Is You Agents, *Proceedings of the 2023 IEEE Congress on Evolutionary Computation*, (pp. 1-8), IEEE.

Lee, C.-S., Wang, M.-H., Chen, C.-Y., Yang, F.-J. Yang, & **Dockhorn, A.** (2023). Genetic Assessment Agent for High-School Student and Machine Co-Learning Model Construction on Computational Intelligence Experience, *Proceedings of the 2023 IEEE Congress on Evolutionary Computation*, IEEE, (pp. 1-8), IEEE.

Wagner, L., Olson, C., & **Dockhorn, A.** (2022). Generalizations of Steering - A Modular Design. In: *Proceedings of the 2022 IEEE Conference on Games*, (pp. 1-4), IEEE.

Xu, L, Hurtado-Grueso, J., Jeurissen, D., Perez-Liebana, D., & **Dockhorn, A.** (2022). Elastic Monte Carlo Tree Search with State Abstraction for Strategy Game Playing. In: *Proceedings of the 2022 IEEE Conference on Games*. (pp. 1-8). IEEE.

Dockhorn, A., Mostaghim, S., Kirst, M., & Zettwitz, M. (2021). Multi-Objective Optimization and Decision-Making in Context Steering. In: *Proceedings of the 2021 IEEE Conference on Games*. (pp. 1-8). IEEE.

Dockhorn, A., Hurtado-Grueso, J., Jeurissen, D., Xu, L, & Perez-Liebana, D., Game State and Action Abstracting Monte Carlo Tree Search for General Strategy Game-Playing, In: *2021 IEEE Conference on Games (CoG)*. (pp. 1-8) IEEE.

Perez-Liebana, D., Guerrero-Romero, C., **Dockhorn, A.**, Xu, L.& Jeurissen, D. (2021). Generating Diverse and Competitive Play-Styles for Strategy Games. In: *Proceedings of the 2021 IEEE Conference on Games*. (pp. 1-8). IEEE.

Dockhorn, A., Hurtado, J., Jeurissen, D., Xu, L., & Pérez-Liébana, D. (2021) Portfolio Search and Optimization for General Strategy Game-Playing. In *Proceedings of the Congress on Evolutionary Computation (CEC)*. (pp. 2085-2092), IEEE. 🏆 **Best Paper Runner-Up Award**

Dockhorn, A., & Kruse, R. (2020). Forward Model Learning for Motion Control Tasks. *Proceedings of the IEEE Intelligent Systems IS'20*, (pp. 1-5). IEEE. 🏆 **Best Paper Award**

Dockhorn, A., & Lucas, S. (2020). Local Forward Model Learning for GVGAI Games. In: *Proceedings of the 2020 IEEE Conference on Games*. (pp. 1-8). IEEE.

Dockhorn, A., Lucas, S. M., Volz, V., Bravi, I., Gaina, R. D., & Pérez-Liébana, D. (2019). Learning Local Forward Models on Unforgiving Games. In: *Proceedings of the 2019 IEEE Conference on Games* (pp. 1–4). IEEE.

Lucas, S. M., **Dockhorn, A.**, Volz, V., Bamford, C., Gaina, R. D., Bravi, I., Pérez-Liébana, D., Mostaghim, S., & Kruse, R. (2019). A Local Approach to Forward Model Learning: Results on the Game of Life Game. In: *Proceedings of the 2019 IEEE Conference on Games*, 1–8. IEEE

Dockhorn, A., Schwensfeier, T., & Kruse, R. (2019). Fuzzy Multiset Clustering for Metagame Analysis. In *Proceedings of the 2019 Conference of the International Fuzzy Systems Association and the European Society for Fuzzy Logic and Technology (EUSFLAT 2019)*, (pp. 1-8). Paris, France: Atlantis Press. 🏆 **Distinguished Student Paper**

Dockhorn, A., Tippelt, T., & Kruse, R. (2018). Model Decomposition for Forward Model Approximation. In *2018 IEEE Symposium Series on Computational Intelligence* (pp. 1751–1757). IEEE.

Dockhorn, A., & Apeldoorn, D. (2018). Forward Model Approximation for General Video Game Learning. In *Proceedings of the 2018 IEEE Conference on Computational Intelligence and Games* (pp. 425–432). IEEE.

Dockhorn, A., Frick, M., Akkaya, Ü., & Kruse, R. (2018). Predicting Opponent Moves for Improving Hearthstone AI. In *17th International Conference on Information Processing and Management of Uncertainty in Knowledge-Based Systems, IPMU 2018* (pp. 621–632). Springer International Publishing.

Sabsch, T., Braune, C., **Dockhorn, A.**, & Kruse, R. (2017). Using a Multiobjective Genetic Algorithm for

Curve Approximation. In *2017 IEEE Symposium Series on Computational Intelligence*, (pp. 1-6). IEEE.

Dockhorn, A., & Kruse, R. (2017). Combining cooperative and adversarial coevolution in the context of pac-man. In *2017 IEEE Conference on Computational Intelligence and Games, CIG 2017* (pp. 60–67). IEEE.

Dockhorn, A., Doell, C., Hewelt, M., & Kruse, R. (2017). A decision heuristic for Monte Carlo tree search doppelkopf agents. In *2017 IEEE Symposium Series on Computational Intelligence* (pp. 1–8). IEEE.

Dockhorn, A., Braune, C., & Kruse, R. (2016). Variable density based clustering. In *2016 IEEE Symposium Series on Computational Intelligence* (pp. 1–8). IEEE.

Dockhorn, A., Braune, C., & Kruse, R. (2015). An Alternating Optimization Approach based on Hierarchical Adaptations of DBSCAN. In *2015 IEEE Symposium Series on Computational Intelligence* (pp. 749–755). IEEE.

Held, P., **Dockhorn, A.**, Krause, B., & Kruse, R. (2015). Clustering Social Networks Using Competing Ant Hives. In *2015 Second European Network Intelligence Conference* (pp. 67–74). IEEE.

Held, P., **Dockhorn, A.**, & Kruse, R. (2014). On Merging and Dividing of Barabasi-Albert-graphs. In *2014 IEEE Symposium on Evolving and Autonomous Learning Systems* (Vol. 444, pp. 17–24).

Held, P., **Dockhorn, A.**, & Kruse, R. (2014). Generating Events for Dynamic Social Network Simulations. *15th International Conference on Information Processing and Management of Uncertainty in Knowledge-Based Systems, IPMU 2014. Communications in Computer and Information Science*, vol 443 (pp. 46–56). Springer, Cham

Workshop Papers (5)

Cook, M., Charity, M., Awiszus, M., Carnovalini, F., & **Dockhorn, A.** (2025). We Call This Controller Skip: AI for Speedrunning. In *Proceedings of the Joint AIIDE Workshops on Experimental AI in Games and Intelligent Narrative Technologies (EXAG-INT 2025)*, CEUR Workshop Proceedings, Vol. 4090, (pp. 1-12).

Xu, L., Pérez-Liébaña, D., & **Dockhorn, A.** (2022). Towards Applicable State Abstractions: a Preview in Strategy Games. In *The Multi-disciplinary Conference on Reinforcement Learning and Decision Making (RLDM) - RL as a Model of Agency Workshop*, (pp. 1-7). RLDM

Dockhorn, A., Jeurissen, D., Hurtado, J., & Pérez-Liébaña, D. (2020). STRATEGA - A General Strategy Games Framework. In *Artificial Intelligence for Strategy Games Decision, AIIDE 2020 Workshop*, (pp. 1-7). AAAI

Gaina, R., Balla, M., **Dockhorn, A.**, Montolio, R., & Pérez-Liébaña, D. (2020). TAG - Tabletop Games Framework. In *Experimental AI in Games (EXAG), AIIDE 2020 Workshop*, (pp. 1-7). AAAI

Dockhorn, A., & Kruse, (2018). Detecting Sensor Dependencies for Building Complementary Model Ensembles. In *Proceedings. 28. Workshop Computational Intelligence*, Dortmund, 29.-30. November 2018 (pp. 217–234).

Preprint Papers (5)

Schmöcker, R., **Dockhorn, A.**, & Rosenhahn, B. (2025). Discovering State Equivalences in UCT Search Trees By Action Pruning. <https://arxiv.org/abs/2510.26346>

Schmöcker, R., **Dockhorn, A.**, & Rosenhahn, B. (2025). Investigating Intra-Abstraction Policies For Non-exact Abstraction Algorithms. <https://arxiv.org/abs/2510.24297>

Pérez-Liébaña, D., **Dockhorn, A.**, Hurtado Gueso, J., & Jeurissen D. (2020). The Design Of "Stratega": A General Strategy Games Framework, (pp. 1–7). <https://arxiv.org/abs/2009.05643>

Gaina, R. D., Balla, M., **Dockhorn, A.**, Montoliu, R., & Pérez-Liébaña D. (2020). Design and Implementation of TAG: A Tabletop Games Framework, (pp. 1–24). <https://arxiv.org/abs/2009.12065>

Dockhorn, A., & Mostaghim, S. (2019). Introducing the Hearthstone-AI Competition, (pp. 1–4). <http://arxiv.org/abs/1906.04238>

Theses and Dissertation (3)

Dockhorn, A. (2020). Dissertation: Prediction-based Search for Autonomous Game-Playing, (pp. 1–231). Otto von Guericke University of Magdeburg. 🏆 **Best Dissertation Award**

Dockhorn, A. (2015). Master Thesis: Hierarchical Extensions and Cluster Validation Techniques for DBSCAN, (pp. 1–101). Otto von Guericke University of Magdeburg.

Dockhorn, A. (2014). Bachelor Thesis: Computergestützte Analyse onkologischer Daten mithilfe Graphischer Modelle, (pp. 1–80). Otto von Guericke University of Magdeburg.

Talks and Presentations

Large Language Models for Behavior Trees Generation in Unreal Engine Foundations of Digital Games (FDG)	Copenhagen 2026
Training a Reinforcement Learning Agent for Tales of Tribute IEEE Conference on Games (COG)	Lisbon 2025
Training a Reinforcement Learning Agent for Tales of Tribute IEEE Conference on Games (COG)	Lisbon 2025
Generating Ensembles of Search Policies to Solve Baba Is You Levels IEEE Conference on Games (COG)	Lisbon 2025
Open-Ended NPC Dialogue Favors Casual Players: A Pilot Comparison of Three LLM-Driven Dialogue Systems IEEE Conference on Games (COG)	Lisbon 2025
Talking to NPCs: Three LLM-Driven Approaches to Dynamic RPG Dialogue IEEE Conference on Games (COG)	Lisbon 2025
Personalized Dynamic Difficulty Adjustment - Imitation Learning Meets Reinforcement Learning IEEE Conference on Games (COG)	Milan 2024
Markov Senior - Learning Markov Junior Grammars to Generate User-specified Content IEEE Conference on Games (COG)	Milan 2024
Keynote — Generating Predictive Models for Unknown Games Workshop on Multimodal Analysis of Spatio-Temporal Data	Hannover (online) 2023
Invited Talk — Inside Games Research: More Than Playing Games USS Insider Event	USS Santiago de Chile (online) 2023
Evolutionary Optimization of “Baba Is You” Agents IEEE Congress on Evolutionary Computation (CEC)	Chicago 2023
Invited Talk — From Specialist to Generalist Game AI AI Grid - Meet Your Mentor	Hannover 2023
Invited Talk — Machine Learning für Modell basierte Planung - Was wir von Game AI lernen können LIFE 2050	Hannover (online) 2022
Generalizations of Steering - A Modular Design IEEE Conference on Games (COG)	Beijing (online) 2022
Keynote — Playing Games to Learn – From Specialist to Generalist AI 2022 IEEE Biennial Congress of Argentina (ARGENCON)	San Juan (online) 2022

Invited Talk — Unity für Steering, Path-Finding und Machine Learning Acagamics e.V.	Magdeburg (online) 2022
Invited Talk — General Strategy Game Playing - A new challenge for AI Manchester Metropolitan University	Manchester (online) 2021
Vorhersagebasierte Suche für autonomes Spielen GI Dissertationspreis	Wadern (online) 2021
Game State and Action Abstracting Monte Carlo Tree Search for General Strategy Game-Playing IEEE Conference on Games (COG)	Copenhagen (online) 2021
Multi-Objective Optimization and Decision-Making in Context Steering IEEE Conference on Games (COG)	Copenhagen (online) 2021
Portfolio Search and Optimization for General Strategy Game-Playing IEEE Congress on Evolutionary Computation (CEC)	Kraków (online) 2021
Keynote — Predictive Search: Algorithms and Applications International Symposium on Signal and Image Processing (ISSIP)	Zagreb (online) 2020
Plenary Lecture — Forward Model Learning for Motion Control Tasks IEEE Intelligent Systems (IS)	Varna (online) 2020
STRATEGA - A General Strategy Games Framework AAAI Conference on Artificial Intelligence and Interactive Digital Entertainment (AIIDE)	Worcester (online) 2020
Local Forward Model Learning for GVGA Games IEEE Conference on Games (COG)	Osaka (online) 2020
Fuzzy Multiset Clustering for Metagame Analysis 11th Conference of the European Society for Fuzzy Logic and Technology (EUSFLAT)	Prague 2019
Learning Local Forward Models on Unforgiving Games 2019 IEEE Conference on Games (COG)	London 2019
Generalisation of Simulation-Based Search for Autonomous Gameplaying Invited Talk - Game AI Research Group	QMUL London 2019
Generalisation of Simulation-Based Search for Autonomous Gameplaying Doktorandentag der Fakultät für Informatik	OVGU Magdeburg 2019
Detecting Sensor Dependencies for Building Complementary Model Ensembles 28. Workshop Computational Intelligence	Dortmund 2018
Forward Model Approximation for General Video Game Learning Conference on Computational Intelligence and Games (CIG)	Maastricht 2018
Predicting Opponent Moves for Improving Hearthstone AI International Conference on Information Processing and Management of Uncertainty in Knowledge-Based Systems (IPMU)	Cádiz 2018
A decision heuristic for Monte Carlo tree search doppelkopf agents 2016 IEEE Symposium Series on Computational Intelligence (SSCI)	Hawaii 2017
Using a Multiobjective Genetic Algorithm for Curve Approximation 2016 IEEE Symposium Series on Computational Intelligence (SSCI)	Hawaii 2017
Combining cooperative and adversarial coevolution in the context of pac-man Conference on Computational Intelligence and Games (CIG)	New York 2017

Variable density based clustering 2016 IEEE Symposium Series on Computational Intelligence (SSCI)	Athen 2016
An Alternating Optimization Approach based on Hierarchical Adaptations of DBSCAN 2015 IEEE Symposium Series on Computational Intelligence (SSCI)	Cape Town 2015
Generating Events for Dynamic Social Network Simulations 2014 International Conference on Information Processing and Management of Uncertainty in Knowledge-Based Systems (IPMU)	Montpellier 2015

Awards and Honours

Outstanding Reviewer Award IEEE Transactions on Games	2026
Best Reviewer Award (1st Runner Up) IEEE Conference on Games 2026	2026
Best Student Paper Award IEEE Conference on Games 2025 for the paper "How do Players Perceive Gender Discrimination? On the Differences of Harassment in Online Games"	2025
APFEL Preis für Exzellente Lehre Leibniz University Hannover, Faculty of Electric Engineering and Computer Science "Award for Best Lecturer of the Year 2023"	2023
Best Paper Runner-Up Award IEEE Congress on Evolutionary Computation 2021 for the paper "Portfolio Search and Optimization for General Strategy Game-Playing"	2021
Best Paper Award IEEE Intelligent Systems 2020 for the paper "Forward Model Learning for Motion Control Tasks"	2020
Best Dissertation Award Department of Computer Science at the Otto von Guericke University	2019/2020
Distinguished Student Paper EUSFLAT 2019 for the paper "Fuzzy Multiset Clustering for Metagame Analysis"	2019
Best Presentation Award Doctoral Symposium at the Otto von Guericke University	2019
Nominee for the "Otto von Guericke Teaching Award" 2018 Otto von Guericke University	2018
Runner-Up Teaching Award "Held der Lehre" for the seminar "Classification Algorithms" Students council of the Department for Computer Science	2017/2018
Runner-Up Teaching Award "Held der Lehre" for the tutorial on "Intelligente Systeme" Students council of the Department for Computer Science	2015/2016
Best Computer Science Master Graduate Department of Computer Science at the Otto von Guericke University	2015/2016

Grants and Sponsored Research Activities

Project (Funding Institution)	Total Funding Volume (k DKK)	Own Share (k DKK)
Novo Nordisk Start Grant Package (NNF)	6492	6492
HybrInt - Hybrid intelligence through interpretable AI in machine perception and interaction (MWK)	22525	1634
Generating Behavior Trees Using Large Language Models (DAAD)	149	75
Scholarship		
"Graduiertenförderung des Landes Sachsen-Anhalt"		2016 – 2018

Commitment

IEEE Computational Intelligence Society (IEEE CIS)

<i>Chair of the IEEE CIS Competition Sub-Committee</i>	2025, 2020-2021
<i>Chair of the IEEE CIS Content Curation Subcommittee</i>	2024
<i>Chair of the IEEE CIS Games Technical Committee</i>	2022-2023
<i>Chair of the IEEE CIS Summer School Sub-Committee</i>	2022-2023
<i>Member of the IEEE CIS Member Activities Committee</i>	2021
<i>Chair of the IEEE CIS Website Sub-Committee</i>	2021-2022
<i>Member of the IEEE CIS Women in Computational Intelligence Sub-Committee</i>	2021-2022
<i>Member of the IEEE CIS Content Curation Sub-Committee</i>	2022-2024
<i>Member of the IEEE CIS Student Activities Sub-Committee</i>	since 2021
<i>Member of the IEEE CIS Education Committee</i>	2020-2025
<i>Member of the IEEE CIS Games Technical Committee</i>	since 2020
<i>Vice-Chair of the IEEE CIS Competition Sub-Committee</i>	2017-2019

Computational Techniques for Tabletop Games Heritage (Game Table) network

<i>Member</i>	2025
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L3S Research Center, Lower Saxony

<i>Member and Research Group Leader</i>	2025
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Otto von Guericke University, Faculty of Computer Science

<i>Member of the Research Commission</i>	2016-2019
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Student Game Developer Club - Acagamics e.V.

<i>Head of Teaching</i>	2017-2018
<i>Head of Industry</i>	2016-2017

Faculty Student Council, Otto von Guericke University, Faculty of Computer Science

<i>Organizer of several events</i>	2013-2015
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Conference, Workshop, and Competition Organization

IEEE Conference on Games (COG) <i>Workshop and Tutorial Chair</i>	2025
IEEE Conference on Games (COG) <i>General Chair</i>	2024
IEEE Conference on Games (COG) <i>Proceedings Chair</i>	2023
IEEE Congress on Evolutionary Computation (CEC) <i>Competition Chair</i>	2023
IEEE World Congress on Computational Intelligence (WCCI) <i>Organizer of the Special Session on Games</i>	2022
IEEE Congress on Evolutionary Computation (CEC) & Genetic and Evolutionary Computation Conference (GECCO) <i>Organizer of the AbstractSwarm Multi-Agent Logistics Competition</i>	2021-2022 (CEC) 2021-2024 (GECCO)
IEEE Symposium Series on Computational Intelligence (SSCI) <i>Organizer of the Interactive Articles and Videos for Education Competition</i>	2021
International Conference on Cognitive and Intelligent Computing (ICCIIC) <i>Member of the Advisory Committee</i>	2021
IEEE Conference on Games (COG) <i>Organizer of the General Strategy Game AI Competition</i>	2021
<i>Tutorial on General Strategy Game AI</i>	2021
IEEE Congress on Evolutionary Computation (CEC) <i>Organizer of the Special Session on Games</i>	2021
IEEE Conference on Games (COG; former IEEE CIG) <i>Organizer of the Hearthstone AI Competition</i>	2018-2020
IEEE Conference on Games (COG; former IEEE CIG) <i>Organizer of the Short Video Competition</i>	2019-2020

Editor and Reviewer Activities

- Journal — Associate Editor.....
- o IEEE Transactions on Games (TOG)
 - o IEEE Computational Intelligence Magazine (CIM) (2024-2025)

Journals — Reviewer.....

- IEEE Computational Intelligence Magazine (CIM)
- IEEE Transactions on Games (TOG)
- IEEE Transactions on Artificial Intelligence (TAI)
- IEEE Access
- ACM Computing Surveys
- Mathematics and Computers in Simulation (MATCOM)
- Transactions on Pattern Analysis and Machine Intelligence
- KI - Künstliche Intelligenz
- Elsevier Information Sciences
- International Journal of Neural Systems (IJNS)
- MDPI Sensors, MDPI Games
- IEEE Transactions on Emerging Topics in Computational Intelligence (TETCI)
- Expert Systems with Applications (ESWA)
- IEEE Transactions on Affective Computing (TAFCC)

Programme Committee Member.....

- The Genetic and Evolutionary Computation Conference (GECCO) 2021
- IEEE Conference on Games (COG), since 2019
- AAAI Conference on Artificial Intelligence and Interactive Digital Entertainment (AIIDE) 2020

Conferences - Reviewer.....

- AAAI Conference on Artificial Intelligence (AAAI), 2024, 2023
- Foundations of Digital Games (FDG), 2023
- IEEE World Congress on Computational Intelligence (WCCI), 2022
- IEEE Conference on Games (COG), 2025, 2024, 2023, 2022, 2021, 2020, 2019
- IEEE Conference on Computational Intelligence in Games (CIG), 2018
- IEEE Symposium Series on Computational Intelligence (SSCI) 2022, 2020, 2016
- The Genetic and Evolutionary Computation Conference (GECCO) 2023, 2022, 2020
- Conference of the European Society for Fuzzy Logic and Technology (Eusflat) 2019
- International Conference on Artificial Intelligence and Soft Computing (ICAISC) 2019
- Conference on Hybrid Intelligent Systems (HIS) 2018
- Conference on Theory and Practice of Natural Computing (TPNC) 2018
- International Symposium on Intelligent Data Analysis (IDA) 2017, 2016
- Australasian Joint Conference on Artificial Intelligence (AI) 2016
- IEEE Intelligent Systems (IS) 2016

Teaching Activities

Lecturer of the SDU Metaverse Lab

<i>Game Programming 2</i>	SS 2026
<i>Introduction to Game AI</i>	SS 2025/2026
<i>Modern Game Development</i>	WS 2026/2027 WS 2025/2026
<i>Game AI in Practice</i>	WS 2026/2027 WS 2025/2026

Lecturer of the Institute for Information Processing

<i>Introduction to Game Development</i>	WS 2023/2024 WS 2022/2023
<i>Graph-based Machine Learning</i>	SS 2024 SS 2023
<i>Seminar: Reinforcement Learning</i>	SS 2022

Appointed Lecturer of the *Visual Computing* working group
Einführung in Digitale Spiele

WS 2023/2024

Appointed Lecturer of the *Computational Intelligence* working group
Computational Intelligence in Games

SS 2018

Teaching Assistant of the *Computational Intelligence* working group
Hot Topics in Computational Intelligence in Games

WS 2021

Fuzzy Systems

SS 2019

Bayes Networks

WS2021/2022

WS 2019/2020

WS 2018/2019

WS 2017/2018

WS 2015/2016

Computational Intelligence in Games

SS 2019

SS 2017

Neural Networks

SS 2018

SS 2015

Seminar Classification Algorithms

WS 2017/2018

Intelligente Systeme

WS 2015/2016

WS 2014/2015

Einführung in die Informatik

WS 2011/2012

(Co-)Supervised Theses

on-going.....

Causal Reinforcement Learning (working title)

PhD Thesis

Mohit Jiwatode

Dynamic Swarm-based Logistics (working title)

Master Thesis

Christian Jørgensen, Marcus Jürgensen

Exploring Continual Learning Methods for Reinforcement Learning Agents in Changing Game Environments (working title)

Master Thesis

Marcus Elmkvist, Toke Odgaard-Jans

2026.....

Adaptive Difficulty Adjustment in draw-based puzzle games

Master Thesis

Mickas Louis Jensen, Lars Philip Francisco Allerup

Exploring using multi-actor simulation to create narratives in games

Bachelor Thesis

August Ellehøj Sepstrup

Online abstractions for Monte Carlo Tree Search

PhD Thesis

Robin Schmöcker

2025.....

Efficient Data Usage for Planning and Reinforcement Learning

PhD Thesis

Linxie Xu

A Comprehensive Study of Early and Late Fusion Strategies in Multimodal Embedding Models <i>Trong Tat Vu</i>	Master Thesis
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2024.....

Dynamic Configuration of Elastic Monte Carlo Tree Search <i>Merle-Sophie Carlotta Wagner</i>	Bachelor Thesis
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Dynamically Adjusting the Use of State Abstractions in Elastic Monte Carlo Tree Search <i>Dominik Schulz</i>	Bachelor Thesis
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Designing Repair Mechanisms to Guarantee Playability of Procedurally Generated Levels <i>Jerrit Gustav Gläsker</i>	Bachelor Thesis
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Analysing the Metagame and Team Dynamics in Competitive MOBA Games <i>Leo Thern</i>	Bachelor Thesis
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Monte Carlo-Baum-Suchverfahren auf der Grundlage eines Graphischen Neuronalen Netzes <i>Aaron Sava</i>	Bachelor Thesis
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Erweiterung des Deep Phase Ansatzes auf verschiedene Eingabemodalitäten <i>Karsten Fuhrberg</i>	Master Thesis
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Short-Term Forecasting of Solar Irradiance with Deep Learning: Application and Enhancement of State-of-the-Art Methods <i>Sebastian Böker</i>	Master Thesis
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Lernen verallgemeinerbarer Zustandsbewertungsheuristiken <i>Erik Albrecht</i>	Bachelor Thesis
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Exploring Generalization and Transfer Learning in AI Agents across Games <i>Malte Hermann</i>	Master Thesis
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Minecraft Architecture Design Using Wave Function Collapse <i>Henrik Bajon</i>	Bachelor Thesis
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Predicting Tire Degradation based on Road Network Analysis <i>Maria Camila Chavez Tobar</i>	Master Thesis
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2023.....

Minecraft Settlement Generation Using Wave Function Collapse <i>Josephine Gisela Helene Klein</i>	Bachelor Thesis 2023
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Dynamically Adjusting the Use of State Abstractions in Elastic Monte Carlo Tree Search <i>Dominik Schulz</i>	Bachelor Thesis 2023
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Neural Architecture Search for Dynamic Difficulty Adjustment of AI Agents <i>Robin Gieseke</i>	Bachelor Thesis 2023
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Optimierung von Portfolio Kompositionen zur Laufzeit <i>Linus Tristan Rattay</i>	Bachelor Thesis 2023
Einbettung von Kontextinformationen in Wave Function Collapse <i>Marian Ryt</i>	Bachelor Thesis 2023
Training and Combining Weak Reinforcement Learners for Enhanced Performance <i>Robin Schmöcker</i>	Master Thesis 2023
Interpreting Latent Vector Representations of World Models <i>Marcel Korneck</i>	Bachelor Thesis 2023
Optimizing Draft Recommendations for MOBA Games Based on General Win Rates and Individual Player Performance <i>Gregor Voigts</i>	Bachelor Thesis 2023
Learning Markov Junior Rule-bases to Generate User-specified Content <i>Mehmet Kayra Oguz</i>	Bachelor Thesis 2023
Dynamic Difficulty Adjustment for Procedural Content Generation in TOAD-GAN <i>Dominic Ludorff</i>	Bachelor Thesis 2023
Development of surrogate models for decentralized energy supply systems <i>Julian Schwarzien</i>	Bachelor Thesis 2023
Designing Repair Mechanisms to Guarantee Playability of Procedurally Generated Levels <i>Jerrit Gustav Gläsker</i>	Bachelor Thesis 2023
Optimizing Deck-drafting Strategies for Collectible Card Game AI using Evolutionary Algorithms <i>Lennart Viet</i>	Bachelor Thesis 2023
Online Optimization of Curriculum Learning Schedules using Evolutionary Optimization <i>Leon Schlecht</i>	Master Thesis 2023
2022	
Driving-AI for Real-Time-Traffic-Simulations <i>Marius Schmidt</i>	Bachelor Thesis 2022
Search-based Procedural Content Generation with Rolling Horizon Evolutionary Algorithm for Tile-based Map Generation <i>Christian Wustrau</i>	Master Thesis 2022
Optimizing Deck-building Strategies for Collectible Card Game AI <i>Can Tuna</i>	Bachelor Thesis 2022
Quality Diversity Optimization for Portfolio-Based Search Algorithms in Real-Time Strategy Games <i>Till Isenhuth</i>	Bachelor Thesis 2022

Context Steering with Differential-Drive Robots: Reactive Navigation based on Multi-Objective Decision-Making	Master Thesis
<i>Nele Traichel</i>	<i>2022</i>

2021.....

Action Abstractions and Real-Time Search for RTS Games (working title)	Master Thesis
<i>Muttahir Mumtaz</i>	<i>2021</i>

Procedural Generation of Rube Goldberg Machines (working title)	Bachelor Thesis
<i>Tomoya Hömberg</i>	<i>2021</i>

2020.....

Neural Network-based Adaption of Rapidly Exploring Random Trees for Motion Planning	Master Thesis
<i>Maximilian Kühn</i>	<i>2020</i>

2019.....

DeePolation: AI-based interpolation on multi-dimensional spherical sensors	Master Thesis
<i>Martin Zettwitz</i>	<i>2019</i>

Clustering of Longitudinal Disease Progression Data	Master Thesis
<i>Aditya Nemali</i>	<i>2019</i>

Evolutionäre Agenten-Generierung für HearthStone	Bachelor Thesis
<i>Bastian Heinrich</i>	<i>2019</i>

Schaffung von Modellierungsansätzen zur Interaktionsvorhersage unbekannter Spiele	Master Thesis
<i>Tim Tippelt</i>	<i>2019</i>

Multivariate Time Series Sensor Data Clustering	Master Thesis
<i>Sourabh Dandage</i>	<i>2019</i>

Prediction of Player Moves in Collectible Card Games	Master Thesis
<i>Tony Schwensfeier</i>	<i>2019</i>

2018.....

Multikriterielle Wegfindung für Agentengruppen	Bachelor Thesis
<i>Maurice Hoffmeister</i>	<i>2018</i>

Classification of Differently Trained Larvae based on Changes in their Trajectories using Artificial Neural Networks	Bachelor Thesis
<i>Jonathan Spiegel</i>	<i>2018</i>

Robust and Transferable Reflectance Reconstruction Using Deep Neural Networks	Master Thesis
<i>Cornelius Styp von Rekowski</i>	<i>2018</i>

Regression Analysis for Power Consumption of a Production Plant	Master Thesis
<i>Sujan Adhikari</i>	<i>2018</i>

Ermitteln der Siegbedingung unbekannter Spiele durch Assoziationsanalyse	Bachelor Thesis
<i>Chris Saxton</i>	<i>2018</i>

Optimising All-Shortest-Path Dictionaries using Machine Learning	Master Thesis
<i>Jannis Becke</i>	<i>2018</i>

Designing an Interface between Data System and Root Cause Evaluation to enhance analysis strategy utilizing a larger Database for Tire Design, Manufacturing and Evaluation Process <i>Pankaj Narula</i>	Master Thesis <i>2018</i>
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2017.....

Entscheidungsheuristische Erweiterung des UCT-Algorithmus für Doppelkopf <i>Matthias Hewell</i>	Master Thesis <i>2017</i>
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Kursverlaufvorhersage von Wertpapieren mit Hilfe von Ensemble Classification <i>Leopold Ryll</i>	Bachelor Thesis <i>2017</i>
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Examples of Press and Publicity

Public presentation of our Game AI Working group Digitales Hannover	Hannover (online) <i>2024</i>
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Interview Guest - Translated Title: "Interview with Alex - How do AI and Gaming Fit Together"	Coding Buddies Podcast <i>2023</i>
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Presentation on Unity for Steering, Path-Finding and Machine Learning Acagamics e.V.	Magdeburg (online) <i>2022</i>
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